

DMSB Course Calendar Project

Documentation

Project Overview

Purpose: To transform class schedule data into a structured, visually clear calendar in Excel that reflects class times, days, and room bookings across DMSB classrooms. The output enables users to filter by room number and view availability.

Request Source: Faculty member needing assistance beyond coding ability, referred by Mark Ferguson, Rohit, and Satish.

Creation Manual (Developer Instructions)

Step 1: Organize Input Data

Start with the provided sheet labeled All. Split necessary columns: Day, Subject, Course Number, Building, Room, Begin, End, Instructor.

Step 2: Combine Course Identifiers

Create a new column to concatenate Subject and Course:

=CONCATENATE(Subject, Course)

Copy this formula throughout the relevant column in the All sheet.

Step 3: Day-Wise Segregation

The dataset spans Monday to Friday. Separate records by day.

Step 4: Time Format Validation

If Begin and End columns are not in Time format, use the Query Editor to convert them.

Step 5: Define Named Ranges

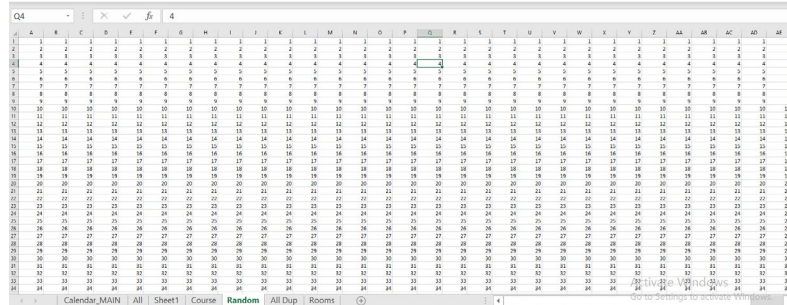
Highlight columns and assign the following names:

- BeginTime → Begin column
- EndTime → End column
- RoomNum → Room column

Use: Formulas > Define Name in Excel

Step 6: Create the "Random" Sheet

Insert a new tab called Random. Generate a list of random integers to serve as lookup indices to the main All sheet.



Step 7: Build Time Matrix

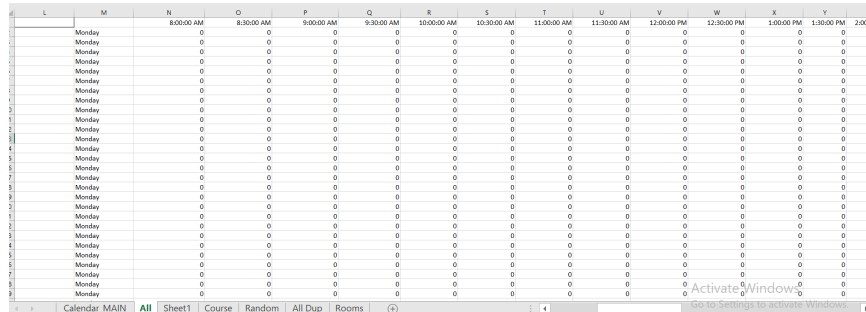
In a new section of the All sheet, list time intervals (30 minutes) from 8:00 AM to 10:00 PM horizontally. Vertically, list each weekday (Monday through Friday).

Step 8: Generate Time Slot Flags

Use the formula to check availability:

=IF(INDEX(RoomNum,Random!A1)=Calendar_MAIN!\$I\$1,

IF(AND(N\$1>=INDEX(BeginTime,Random!A1),N\$1<=INDEX(EndTime,Random!A1)),1,0),0)



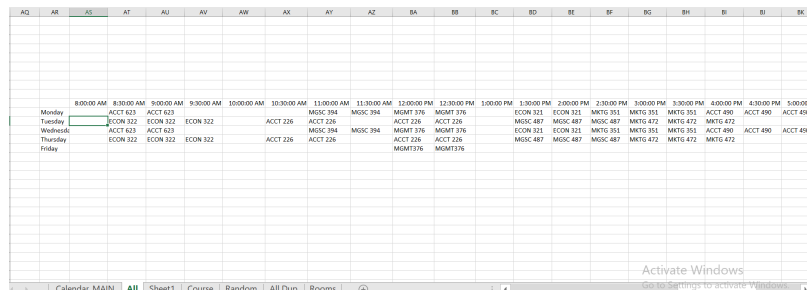
Step 9: Populate Class Info by Day

For each weekday, use:

=IF(SUMIFS(N2:N1107,\$M\$2:\$M\$1107,"Monday")=0,"",

INDEX(\$A\$2:\$J\$1107,MATCH(1,(\$M\$2:\$M\$1107="Monday")*(N2:N1107=1),0),4))

Update the day string ("Monday") accordingly for other weekdays.



Step 10: Transpose to Calendar_MAIN

Use:

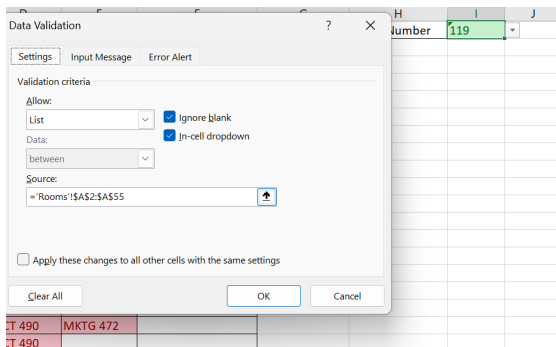
=TRANSPOSE(All!AR10:BU15)

Place this in the Calendar_MAIN sheet.

	A	B	C	D	E	F	G	H	I
1		Monday	Tuesday	Wednesday	Thursday	Friday		Room Number	119
2	8:00:00 AM								
3	8:30:00 AM	ACCT 623	ECON 322	ACCT 623	ECON 322				
4	9:00:00 AM	ACCT 623	ECON 322	ACCT 623	ECON 322				
5	9:30:00 AM		ECON 322		ECON 322				
6	10:00:00 AM								
7	10:30:00 AM		ACCT 226		ACCT 226				
8	11:00:00 AM	MGSC 394	ACCT 226	MGSC 394	ACCT 226				
9	11:30:00 AM	MGSC 394		MGSC 394					
10	12:00:00 PM	MGMT 376	ACCT 226	MGMT 376	ACCT 226	MGMT376			
11	12:30:00 PM	MGMT 376	ACCT 226	MGMT 376	ACCT 226	MGMT376			
12	1:00:00 PM								
13	1:30:00 PM	ECON 321	MGSC 487	ECON 321	MGSC 487				
14	2:00:00 PM	ECON 321	MGSC 487	ECON 321	MGSC 487				
15	2:30:00 PM	MKTG 351	MGSC 487	MKTG 351	MGSC 487				
16	3:00:00 PM	MKTG 351	MKTG 472	MKTG 351	MKTG 472				
17	3:30:00 PM	MKTG 351	MKTG 472	MKTG 351	MKTG 472				
18	4:00:00 PM	ACCT 490	MKTG 472	ACCT 490	MKTG 472				
19	4:30:00 PM	ACCT 490		ACCT 490					
20	5:00:00 PM	ACCT 490		ACCT 490					
21	5:30:00 PM								

Step 11: Setup Room Dropdown

Create a tab labeled Rooms. List all unique room numbers and sort them. In cell I1 of Calendar_MAIN, go to Data > Data Validation → Allow: List
Source: =Rooms!A1:A[n] (adjust n to final row)



Step 12: Apply Conditional Formatting

Go to Home > Conditional Formatting. Apply rule to highlight non-blank cells in red to indicate occupied time slots.

Step 13: Final Test

Ensure selecting a room number in Calendar_MAIN updates the calendar dynamically.

User Manual (For Calendar Users)

Step 1: Select Room

In Calendar_MAIN, cell I1, use dropdown to select a Room Number (e.g., 119).

Step 2: View Calendar

View the calendar grid for that room:

- Red/filled cells = Time slots with classes.
- Blank cells = Available for booking.

	A	B	C	D	E	F	G	H	I
1		Monday	Tuesday	Wednesday	Thursday	Friday		Room Number	119
2	8:00:00 AM								
3	8:30:00 AM	ACCT 623	ECON 322	ACCT 623	ECON 322				
4	9:00:00 AM	ACCT 623	ECON 322	ACCT 623	ECON 322				
5	9:30:00 AM		ECON 322		ECON 322				
6	10:00:00 AM								
7	10:30:00 AM		ACCT 226		ACCT 226				
8	11:00:00 AM	MGSC 394	ACCT 226	MGSC 394	ACCT 226				
9	11:30:00 AM	MGSC 394		MGSC 394					
10	12:00:00 PM	MGMT 376	ACCT 226	MGMT 376	ACCT 226	MGMT376			
11	12:30:00 PM	MGMT 376	ACCT 226	MGMT 376	ACCT 226	MGMT376			
12	1:00:00 PM								
13	1:30:00 PM	ECON 321	MGSC 487	ECON 321	MGSC 487				
14	2:00:00 PM	ECON 321	MGSC 487	ECON 321	MGSC 487				
15	2:30:00 PM	MKTG 351	MGSC 487	MKTG 351	MGSC 487				
16	3:00:00 PM	MKTG 351	MKTG 472	MKTG 351	MKTG 472				
17	3:30:00 PM	MKTG 351	MKTG 472	MKTG 351	MKTG 472				
18	4:00:00 PM	ACCT 490	MKTG 472	ACCT 490	MKTG 472				
19	4:30:00 PM	ACCT 490		ACCT 490					
20	5:00:00 PM	ACCT 490		ACCT 490					
21	5:30:00 PM								

Calendar_MAIN All Sheet1 Course Random All Dup Rooms +



DMSB Calendar.xlsx